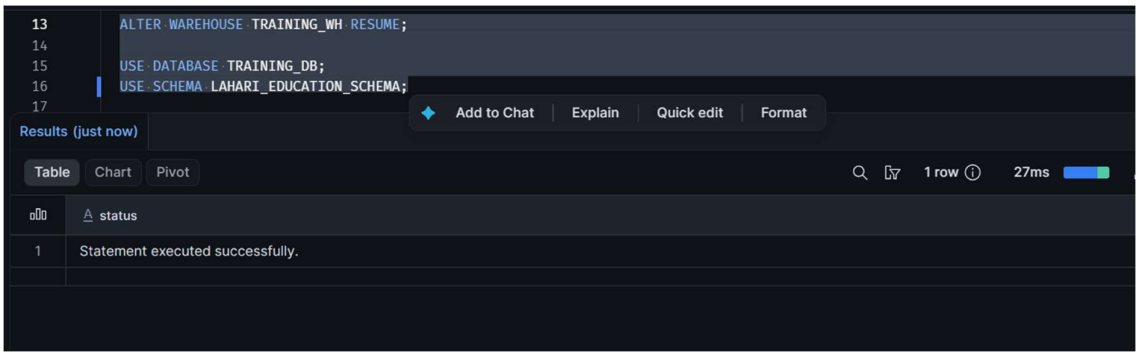
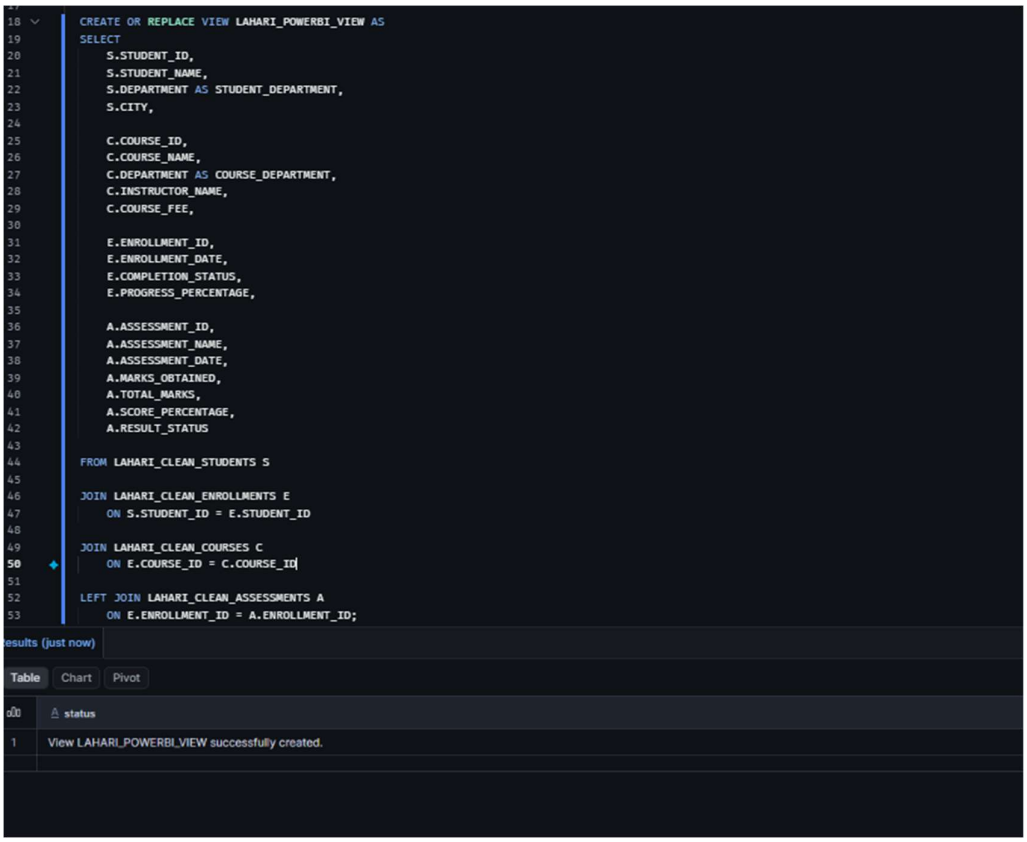
POWER BI + SNOWFLAKE VISUALIZATION

Using the database, schema and warehouse



Creating a view using the existing University Learning Analytics Dataset in snowflake. It contains students, courses, enrollments, and assessments, so you can create a dashboard with multiple KPIs and charts.



Verify:

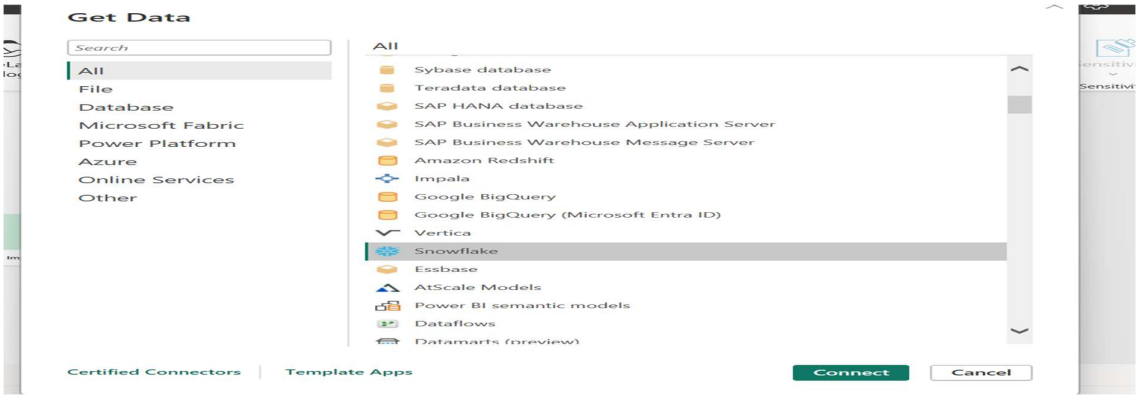
```
56  SELECT * FROM LAHARI_POWERBI_VIEW;
```

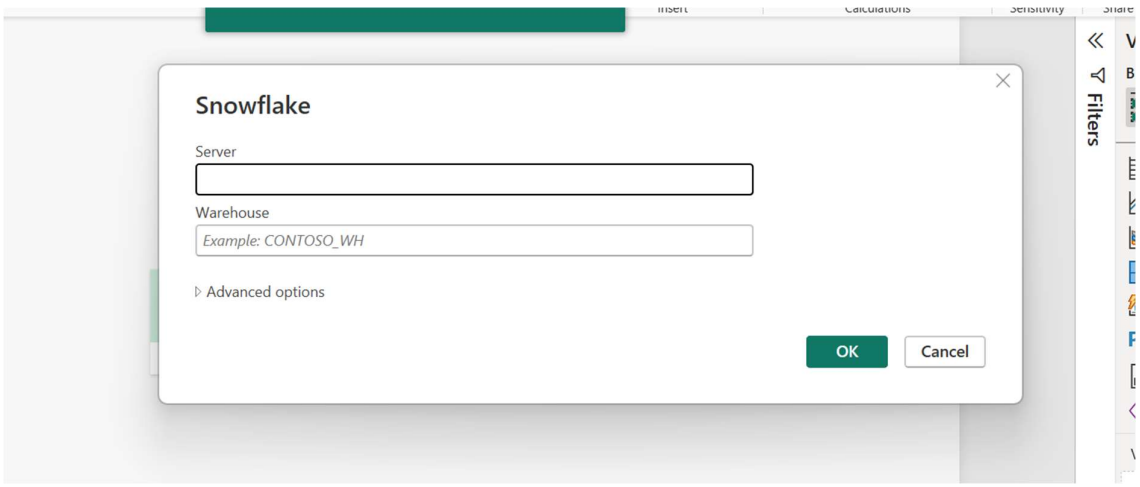
Results (just now)

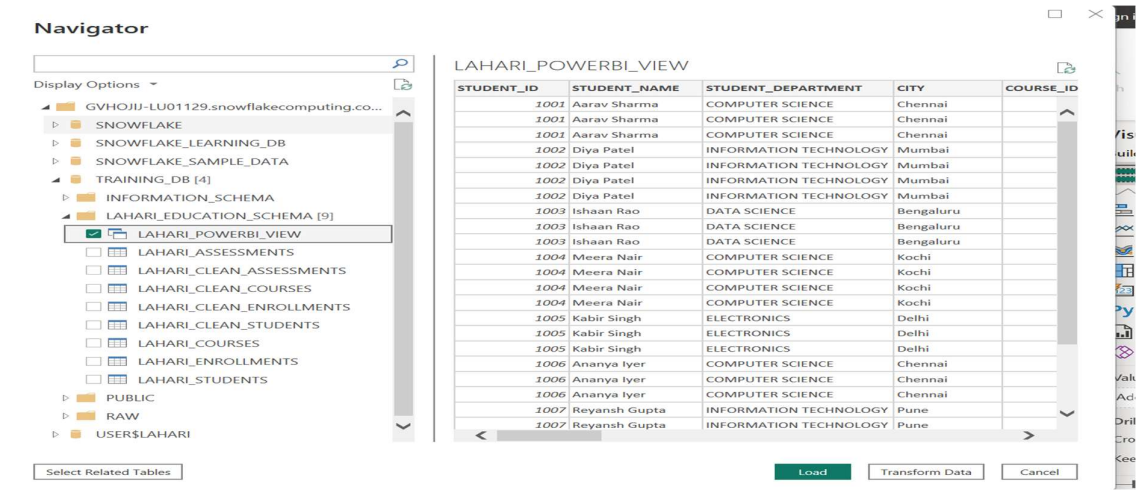
Table | Chart | Pivot — 100 rows — 848ms

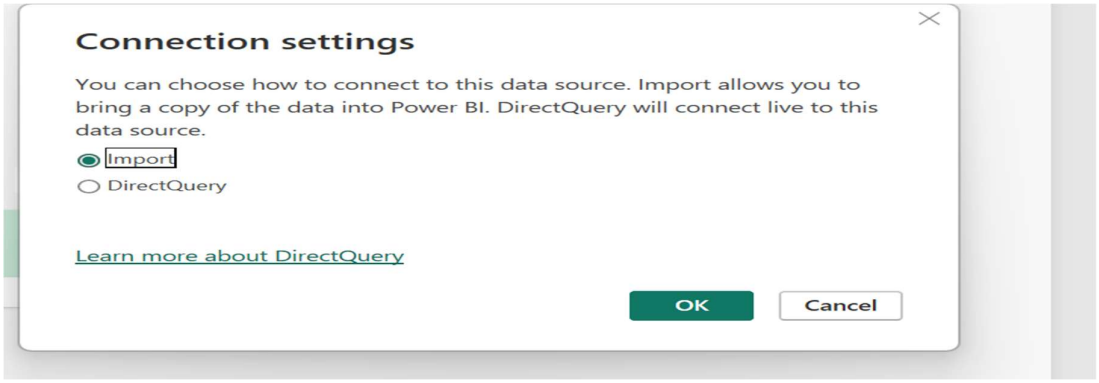
	STUDENT_ID (1001–1030)	STUDENT (Aadhya Men... 4.0%, Atharv Verma 4.0%, +28 more)	STUDENT_DEPARTMENT (COMPUTER SCIENCE 22.0%, INFORMATION TEC... 22.0%, +3 more)	CITY (Ch... 12.0%, Be... 11.0%, +12 more)	COURSE_ID (201–210)	COURSE_NAME (Digital Electro... 12.0%, Machine Learn... 12.0%, +8 more)	COURSE_DEPARTMENT (COMPUTER SCIEN... 28.0%, DATA SCIENCE 22.0%, +3 more)	INSTRUCTOR_NAME (Dr. Nisha Menon 12.0%, Prof. Sanjay Me... 12.0%, +8 more)
1	1001	Aarav Sharma	COMPUTER SCIENCE	Chennai	201	Python Programming	COMPUTER SCIENCE	Dr. Kavitha Rao
2	1001	Aarav Sharma	COMPUTER SCIENCE	Chennai	203	Machine Learning	DATA SCIENCE	Dr. Nisha Menon
3	1001	Aarav Sharma	COMPUTER SCIENCE	Chennai	203	Machine Learning	DATA SCIENCE	Dr. Nisha Menon
4	1002	Diya Patel	INFORMATION TECHNOLOGY	Mumbai	202	Database Management	INFORMATION TECHNOLOGY	Prof. Arvind Kumar
5	1002	Diya Patel	INFORMATION TECHNOLOGY	Mumbai	202	Database Management	INFORMATION TECHNOLOGY	Prof. Arvind Kumar
6	1002	Diya Patel	INFORMATION TECHNOLOGY	Mumbai	206	Cybersecurity Basics	COMPUTER SCIENCE	Prof. Vikram Das
7	1002	Diya Patel	INFORMATION TECHNOLOGY	Mumbai	206	Cybersecurity Basics	COMPUTER SCIENCE	Prof. Vikram Das
8	1003	Ishaan Rao	DATA SCIENCE	Bengaluru	203	Machine Learning	DATA SCIENCE	Dr. Nisha Menon
9	1003	Ishaan Rao	DATA SCIENCE	Bengaluru	203	Machine Learning	DATA SCIENCE	Dr. Nisha Menon

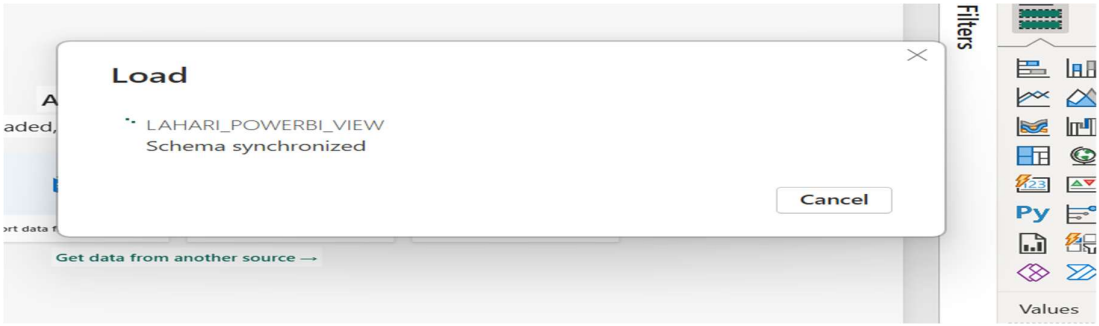
Connect Power BI to Snowflake











Create DAX Measures

In Power BI, select Modeling → New measure.

Total Students

```
1 Total Students =
2 DISTINCTCOUNT(LAHARI_POWERBI_VIEW[STUDENT_ID])
```

Total Enrollments

```
1 Total Enrollments =
2 DISTINCTCOUNT(LAHARI_POWERBI_VIEW[ENROLLMENT_ID])
```

Completed Enrollments

```
1 Completed Enrollments =
2 CALCULATE(
3     DISTINCTCOUNT(LAHARI_POWERBI_VIEW[ENROLLMENT_ID]),
4     LAHARI_POWERBI_VIEW[COMPLETION_STATUS] = "COMPLETED"
5 )
```

Completion Rate

```
1 Completion Rate =
2 DIVIDE(
3     [Completed Enrollments],
4     [Total Enrollments],
5     0
6 )
```

Average Progress

```
1 Average Progress =
2 AVERAGE(LAHARI_POWERBI_VIEW[PROGRESS_PERCENTAGE])
```

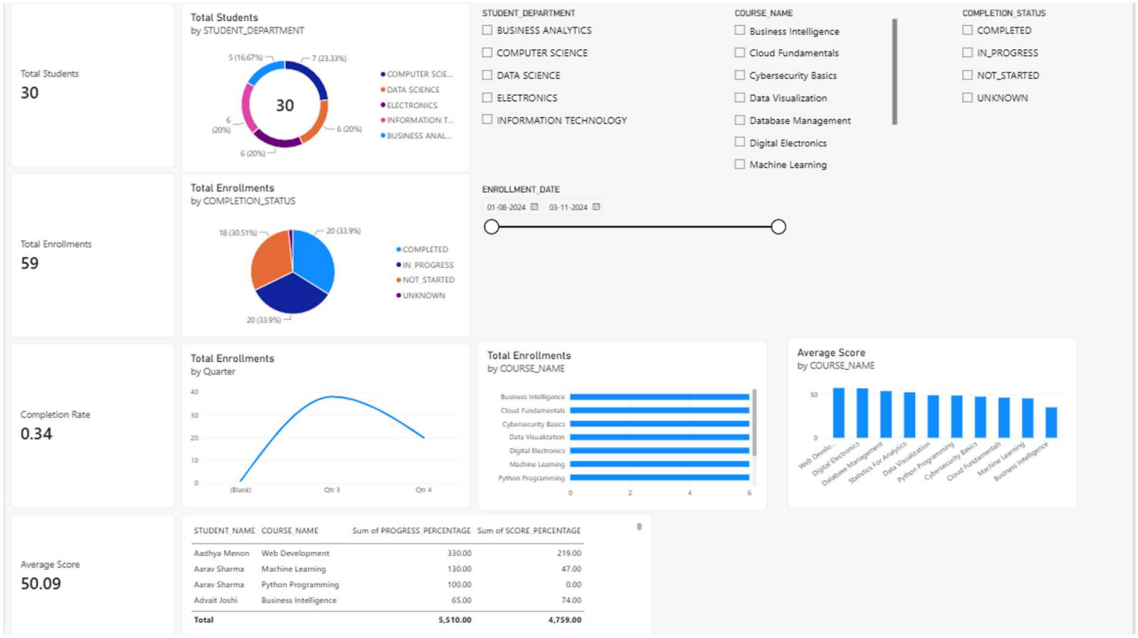
Average Assessment Score

```
1 Average Score =
2 AVERAGE(LAHARI_POWERBI_VIEW[SCORE_PERCENTAGE])
```

Total Courses

```
1 Total Courses =
2 DISTINCTCOUNT(LAHARI_POWERBI_VIEW[COURSE_ID])
```

Final Dashboard:



Visual	**Fields**
Card	Total Students
Card	Total Enrollments
Card	Completion Rate
Card	Average Score
Clustered bar chart	Course Name → Total Enrollments
Donut chart	Student Department → Total Students
Column chart	Course Name → Average Score
Pie chart	Completion Status → Total Enrollments
Line chart	Enrollment Date → Total Enrollments
Table	Student Name, Course Name, Progress Percentage, Score Percentage